

Analisi II per Ingegneria Gestionale,
Appello 28 Giugno 2022

Esercizio I:[4 pt]

Calcolare

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(xy^5)}{xy \ln(1 + x^2 + y^4)}$$

Esercizio II:(pt 6)

Calcolare

$$\iint_A x^2 dx dy$$

dove $A = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 2x, y \geq x\}$.

Esercizio III: (pt 6)

Calcolare il lavoro del campo vettoriale $F = (x^2, xy)$ lungo la curva chiusa percorsa in senso antiorario descritta dall'equazione cartesiana $x^2 + y^2 - y = 0$.

Esercizio IV: (pt 10)

Dato il campo vettoriale $F = (xy, xy, z^2)$, calcolarne il flusso attraverso la superficie laterale del tronco di cono descritto dall'equazione $z^2 = x^2 + y^2$, $z \in (1, \sqrt{2})$.

Esercizio V: (pt 4)

Data la funzione $f(x, y) = x^2 + y^2 - x - y$ determinare il valore di massimo e minimo assoluto nell'insieme

$$A = \{(x, y) \in \mathbb{R}^2, |x| \leq 2, |y - 1| \leq 2\}.$$

Esercizio I: [4 pt]

Calcolare

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(xy^5)}{xy \ln(1+x^2+y^4)}$$

$$\begin{aligned} \lim_{\rho, \theta \rightarrow 0} \frac{\rho^6 \cos \theta \sin^5 \theta}{xy (x^2 + y^4)} &= \frac{\rho^6 \cos \theta \sin^5 \theta}{\rho^2 \cos \theta \sin \theta (\rho^2 \cos^2 \theta + \rho^4 \sin^4 \theta)} \\ &= \frac{\cancel{\rho^6} \cancel{\cos \theta} \sin^5 \theta}{\cancel{\rho^4} \cancel{\cos \theta} \sin \theta (\cos^2 \theta + \rho^2 \sin^4 \theta)} = 0 \end{aligned}$$

Esercizio II: (pt 6)

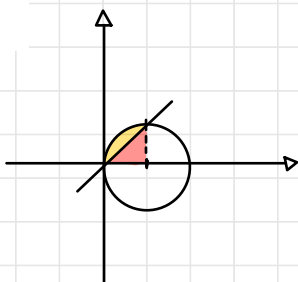
Calcolare

$$\iint_A x^2 dx dy$$

dove $A = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 2x, y \geq x\}$.

Studiamo il dominio:

$$A: (x, y) \in \mathbb{R}^2 \quad x^2 + y^2 - 2x \leq 0 \quad \forall y \geq x$$



$$A = A_{\text{quarto di arco}} - A_{\text{triangolo}}$$

$$A_{\text{quarto di arco}} = \int_{\frac{\pi}{2}}^{\pi} \int_0^1 \rho^2 \cos^2 \theta \, \rho \, d\rho \, d\theta = \int_{\frac{\pi}{2}}^{\pi} \int_0^1 \rho^3 \cos^2 \theta \, d\rho \, d\theta = \int_0^1 \rho^3 \left| \frac{1}{2} (x + \sin x \cos x) \right|_{\frac{\pi}{2}}^{\pi} d\rho$$

$$= \frac{1}{2} \int_0^1 \rho^3 \left| \pi - \frac{\pi}{2} \right| d\rho = \frac{\pi}{16}$$

$$A_{\text{triangolo}} = \{x, y \in \mathbb{R}^2 : 0 < x < 1, 0 < y < x\}$$

$$\int_0^1 x^2 \int_0^x dy \, dx = \frac{1}{4}$$

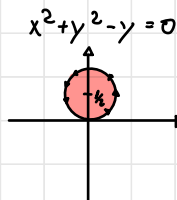
$$A_{\text{rec}} = \frac{\pi}{16} - \frac{1}{4} = \frac{\pi - 4}{16}$$

Esercizio III: (pt 6)

Calcolare il lavoro del campo vettoriale $F = (x^2, xy)$ lungo la curva chiusa percorsa in senso antiorario descritta dall'equazione cartesiana $x^2 + y^2 - y = 0$.

$$F = (x^2, xy) \quad \text{curva chiusa, senso orario}$$

$$\begin{aligned} \text{rot}(F) &= \frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} = y - 0 \\ &= y \quad \square \text{ Quindi } F \text{ non è} \\ &\quad \text{conservativo} \end{aligned}$$



$$\begin{aligned} y &= \cos t; \quad \frac{1}{2} + \sin t \\ \dot{y} &= -\sin t, \quad \cos t \end{aligned}$$

Uso il th del rotore $L = \iint_E y \, dx \, dy$, $\int_0^{2\pi} \int_0^{\frac{1}{2}} (\frac{1}{2} + \sin \theta) p \, dp \, d\theta$

$$\int_0^{2\pi} \int_0^{\frac{1}{2}} \frac{1}{2} p + p \sin \theta \, dp \, d\theta$$

$$\int_0^{2\pi} \int_0^{\frac{1}{2}} \frac{1}{2} p \, dp \, d\theta + \int_0^{2\pi} \int_0^{\frac{1}{2}} p \sin \theta \, dp \, d\theta$$

$$\frac{1}{4} \int_0^{2\pi} \left[p^2 \right]_0^{\frac{1}{2}} d\theta + 0$$

$$\frac{1}{4} \int_0^{2\pi} \frac{1}{4} d\theta = \frac{1}{16} 2\pi = \frac{\pi}{8} \checkmark$$

Posso farlo con la def

$$L = \int_0^{2\pi} [\cos^2 t, \cos t (\frac{1}{2} + \sin t)] [-\sin t, \cos t] dt$$

$$= \int_0^{2\pi} [-\sin t \cos^2 t + \cos^2 t (\frac{1}{2} + \sin t)] dt$$

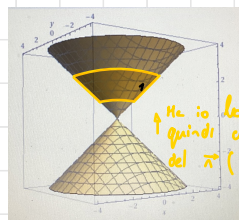
$$= \int_0^{2\pi} \cos^2 t (-\cancel{\sin t} + \frac{1}{2} + \cancel{\sin t}) dt = \frac{1}{2} \left| \frac{1}{2} (x + \sin x \cos x) \right|_0^{2\pi}$$

$$= \frac{1}{2} \left| \frac{1}{2} (2\pi - 0) \right|$$

$$= \frac{\pi}{2}$$

Esercizio IV: (pt 10)

Dato il campo vettoriale $F = (xy, xy, z^2)$, calcolarne il flusso attraverso la superficie laterale del tronco di cono descritto dall'equazione $z^2 = x^2 + y^2$, $z \in (1, \sqrt{2})$.



$$F = (xy, xy, z^2) \quad z^2 = x^2 + y^2, \quad z \in (1, \sqrt{2})$$

$$\operatorname{div} F = (y + x + 2z)$$

una possibile opzione è calcolare il flusso con la div
l'altra è usare la def.

$$\phi(F, \mathcal{G}) = \int \vec{F} \cdot \vec{n} \, d\mathcal{G}$$

$$\text{dove } \vec{n} = \det(J)$$

Devo però trovare una parametrizzazione per γ : $\gamma = (x, y, \sqrt{x^2 + y^2})$

$$\text{adesso calcolo } \frac{\partial \gamma}{\partial x} = 1, 0, \frac{x}{\sqrt{x^2 + y^2}} = 1, 0, \frac{x}{\sqrt{x^2 + y^2}}$$

$$\frac{\partial \gamma}{\partial y} = 0, 1, \frac{y}{\sqrt{x^2 + y^2}} = 0, 1, \frac{y}{\sqrt{x^2 + y^2}}$$

$$J = \begin{vmatrix} i & j & k \\ 1 & 0 & \frac{x}{\sqrt{x^2 + y^2}} \\ 0 & 1 & \frac{y}{\sqrt{x^2 + y^2}} \end{vmatrix}$$

$$\det(J) = -\frac{x}{\sqrt{x^2 + y^2}}; -\frac{y}{\sqrt{x^2 + y^2}}; 1$$

F avrebbe z^2 che però è $x^2 + y^2$

$$\phi = \int \vec{F} \cdot \vec{n} \, d\mathcal{G} = \int (xy, xy, x^2 + y^2) \left(+\frac{x}{\sqrt{x^2 + y^2}} i + \frac{y}{\sqrt{x^2 + y^2}} j - 1 \right)$$

$$= \iint \frac{x^2 y}{\sqrt{x^2 + y^2}} + \frac{xy^2}{\sqrt{x^2 + y^2}} - x^2 - y^2 \, dx \, dy$$

$$\text{con } 1 < x^2 + y^2 < 2$$

↳ prendo questa

Esercizio V: (pt 4)

Data la funzione $f(x, y) = x^2 + y^2 - x - y$ determinare il valore di massimo e minimo assoluto nell'insieme

$$A = \{(x, y) \in \mathbb{R}^2, |x| \leq 2, |y - 1| \leq 2\}.$$

Parto dallo studiare l'insieme A

$$A = \{(x, y) \in \mathbb{R}^2, |x| \leq 2, |y - 1| \leq 2\}$$

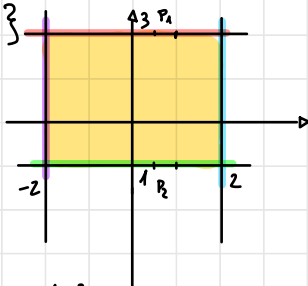
$$\text{se } x > 0, \quad x < 2$$

$$x < 0, \quad -x < 2, \quad x > -2$$

$$|y - 1| < 2$$

$$\text{se } y - 1 > 0, \quad y > 1 \quad \text{allora} \quad \begin{matrix} y - 1 < 2 \\ y < 3 \end{matrix}$$

$$\begin{matrix} y < 1 \quad \text{allora} \quad -y + 1 < 2 \\ -y < 1 \\ y > -1 \end{matrix}$$



Quindi sto valutando la funzione in un quadrato.

Studio normalmente la funzione:

$$f(x, y) = x^2 + y^2 - x - y$$

$$\nabla f = \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$$

$$\frac{\partial f}{\partial x} = 2x - 1 \begin{cases} \frac{\partial^2 f}{\partial x \partial x} = 2 \\ \frac{\partial^2 f}{\partial x \partial y} = 0 \end{cases}$$

$$\frac{\partial f}{\partial y} = 2y - 1 \begin{cases} \frac{\partial^2 f}{\partial y \partial y} = 2 \\ \frac{\partial^2 f}{\partial y \partial x} = 0 \end{cases}$$

$$H = \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix}$$

quindi $\det(H) = 4$
 $\rightarrow$ Quindi ho un minimo

$$\begin{cases} 2x - 1 = 0 & x = \frac{1}{2} \\ 2y - 1 = 0 & y = \frac{1}{2} \end{cases}$$

quindi il punto $P(\frac{1}{2}; \frac{1}{2})$ = punto di minimo

$$f(P) = -\frac{1}{2}$$

Adesso studio A lungo i bordi essendo retta angolare

$$\begin{aligned} f(y=3) &= x^2 + 9 - x - 3 \\ &= x^2 - x + 6 \end{aligned} \quad \text{quindi a questo è analisi in una variabile sola}$$

$$\frac{\partial f}{\partial x} = 2x - 1 \quad x = \frac{1}{2} \quad f\left(\frac{1}{2}, 3\right) = \frac{1}{4} + 9 - \frac{1}{2} - 3 = \frac{23}{4}$$

$$\begin{aligned} f(y=-1) &= x^2 + 1 - x + 1 \\ &= x^2 - x + 2 \end{aligned}$$

$$\frac{\partial f}{\partial x} = 2x - 1 \quad x = \frac{1}{2} \quad f\left(\frac{1}{2}, -1\right) = \frac{1}{4} + 1 - \frac{1}{2} + 1 = \frac{7}{4}$$

$$\begin{aligned} f(x=2) &= 4 + y^2 - 2 - y \\ &= y^2 - y + 2 \end{aligned}$$

$$\frac{\partial f}{\partial y} = 2y - 1 \quad y = \frac{1}{2} \quad P_3\left(2, \frac{1}{2}\right) = \frac{5}{4}$$

$$\begin{aligned} f(x=-2) &= 4 + y^2 + 2 - y \\ &= y^2 - y + 6 = 0 \end{aligned}$$

$$\frac{\partial f}{\partial y} = 2y - 1 \quad y = \frac{1}{2} \quad P_4\left(-2, \frac{1}{2}\right) = \frac{21}{4}$$

$$\text{Quindi ho } P\left(\frac{1}{2}, \frac{1}{2}\right) = \text{MINIMO}$$

$$P\left(\frac{1}{2}, 3\right) = \text{MAX}$$

1° APPELLO 2022

$$① \iint_{\Sigma} (x+z) d\sigma \quad \Sigma = \{2x+3y+6z=9\} \quad z = \frac{3}{2} - \frac{1}{2}y - \frac{1}{3}x$$

$$d\sigma = \sqrt{1 + \left(-\frac{1}{2}\right)^2 + \left(-\frac{1}{3}\right)^2} = \frac{7}{6}$$

$$\frac{7}{6} \iint_{\Sigma} x + \frac{3}{2} - \frac{1}{2}y - \frac{1}{3}x d\alpha d\gamma = \frac{7}{6} \iint_{\Sigma} \frac{2}{3}x - \frac{1}{2}y + \frac{3}{2} d\alpha d\gamma =$$

$$z=0 \rightarrow 2x+3y=9 \rightarrow y = 3 - \frac{2}{3}x$$



$$\int_0^{\frac{3}{2}} \int_0^{3-\frac{2}{3}x} \left(\frac{2}{3}x - \frac{1}{2}y + \frac{3}{2} \right) d\alpha d\gamma = \int_0^{\frac{3}{2}} \left| \frac{2}{3}x\gamma - \frac{1}{4}\gamma^2 + \frac{3}{2}\gamma \right|_0^{3-\frac{2}{3}x} d\alpha =$$

$$= \int_0^{\frac{3}{2}} \left(\frac{2}{3}x \left(3 - \frac{2}{3}x \right) - \frac{1}{4} \left(3 - \frac{2}{3}x \right)^2 + \frac{3}{2} \left(3 - \frac{2}{3}x \right) \right) d\alpha =$$

$$= \frac{7}{6} \int_0^{\frac{3}{2}} \left(2x - \frac{4}{9}x^2 - \frac{9}{4} + \frac{1}{9}x^2 + \frac{9}{2} - x \right) d\alpha = \frac{7}{6} \int_0^{\frac{3}{2}} \left(x - \frac{5}{9}x^2 + \frac{9}{4} \right) d\alpha =$$

$$= \frac{7}{6} \left| \frac{x^2}{2} - \frac{5}{27}x^3 + \frac{9}{4}x \right|_0^{\frac{3}{2}} = \frac{7}{6} \cdot \left(\frac{81}{4} - \frac{135}{8} + \frac{81}{8} \right) = \frac{7}{6} \cdot \frac{108}{8} = \frac{63}{4}$$

6
2

$$\iint_A \left(\frac{1}{xy} \right) dx dy \quad A = \left\{ 0 < x < y < 3x, 1 < x+y < 2 \right\}$$

$$\begin{cases} \frac{y}{x} = m \\ x + y = v \end{cases} \begin{cases} y = m(v - y) \\ x = v - y \end{cases} \begin{cases} y = mv - my \\ y = \frac{mv}{1+m} \\ x = \frac{v}{1+m} \end{cases}$$

$$J = \begin{pmatrix} \frac{-v}{(1+n)^2} & \frac{1}{(1+n)} \\ \frac{v}{(1+n)^2} & \frac{n}{(1+n)} \end{pmatrix} = \frac{-v \cdot n - v}{(1+n)^3} = \frac{v}{(1+n)^2}$$

$$\iint_A \frac{1}{xy} dx dm = \iint_E \frac{(1+m)^2}{m v^2} \cdot \frac{v}{(1+m)^2} du dv \quad E = \{(m, v) \in \mathbb{R}^2 \mid 1 \leq m \leq 3, 1 \leq v \leq 2\}$$

$$\int_1^2 \int_1^3 \frac{1}{u \cdot v} du dv = \log(3) \cdot \log(2)$$

$$\text{rot} = x^2 + 2 - x^2 = 2$$

$$2 \int_0^{\frac{\pi}{2}} \int_0^{\cos \theta} p \, dp \, d\theta = 2 \int_0^{\frac{\pi}{2}} \frac{(\cos \theta)^2}{2} = \frac{\pi}{4} + 0 = \frac{\pi}{4}$$

$$5) f(x,y) = x^2y - 2x^2 - y^2 \quad \nabla f = (2xy - 4x, x^2 - 2y)$$

$$f_{xx} = 2y - 4 \quad f_{yy} = -2 \quad f_{xy} = 2x \quad f_{yx} = 2x$$

$$\begin{cases} 2x(y-2) = 0 \\ x^2 = 2y \end{cases} \begin{cases} A(0,0) & B(2,2) \\ C(-2,2) \end{cases} \quad \nabla^2 f(x,y) = \begin{pmatrix} 2(y-2) & 2x \\ 2x & -2 \end{pmatrix}$$

$B \in C$ $\det < 0$ saddle $A \rightarrow \det > 0, f_{xx} < 0$: max

↳ 3

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} = 1 \quad y = (1 - x^{\frac{2}{3}})^{\frac{3}{2}} \quad y' = -\frac{2}{3} \cdot x^{-\frac{1}{3}} \cdot (1 - x^{\frac{2}{3}})^{\frac{1}{2}} \cdot \frac{3}{2}$$

$$(\dot{y})^2 = x^{-\frac{2}{3}} (1 - x^{\frac{2}{3}}) = x^{-\frac{2}{3}} - 1 \quad \sqrt{1 + f'^2} = x^{-\frac{1}{3}}$$

$$2 \int_0^1 x^{-\frac{1}{3}} = 3$$